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1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  """
5  Assignment 5
6  Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be differentiable. To minimize  $f$ , consider the gradient descent
7   $x_{(n+1)} = x_n - \eta * f'(x_n)$ .
8  """
9
10
11 # Chosen convex function:  $f(x) = (x - 3)^2$ 
12 def f_convex(x): 2 usages
13     return (x - 3) ** 2
14
15 def f_derivative_convex(x): 1 usage
16     return 2 * (x - 3)
17
18
19 # Non-convex function example
20 # Use  $f(x) = x^4 - 4x^2 + 3$  (has 2 minima at  $\pm\sqrt{2}$ )
21 def f_non_convex(x): 2 usages
22     return x**4 - 4*x**2 + 3
23
24 def f_non_convex_derivative(x): 1 usage
25     return 4*x**3 - 8*x

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29 # Gradient descent functions
30 def gradient_descent_convex(rate, initial_x, iters): 3 usages
31     x = initial_x
32     trajectory = [x]
33
34     for i in range(iters):
35         x = x - rate * f_derivative_convex(x)
36         trajectory.append(x)
37
38     return x, trajectory
39
40
41 def gradient_descent_non_convex(rate, initial_x, iters): 1 usage
42     x = initial_x
43     trajectory = [x]
44
45     for i in range(iters):
46         x = x - rate * f_non_convex_derivative(x)
47         trajectory.append(x)
48
49     return x, trajectory
50

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53 # Graphing (convex case)
54 def graph_function_convex(minimizer, trajectory, rate): 3 usages
55     x_vals = np.linspace(-2, stop: 6, num: 400)
56     y_vals = f_convex(x_vals)
57
58     plt.plot(*args: x_vals, y_vals, label='f(x) = (x - 3)^2')
59     traj_x = np.array(trajectory)
60     traj_y = f_convex(traj_x)
61
62     plt.scatter(traj_x, traj_y, color='red', marker='x',
63                label='Gradient Descent Path')
64
65     plt.xlabel('x')
66     plt.ylabel('f(x)')
67     plt.legend()
68     plt.title(f'Convex Function - Gradient Descent (n = {rate})')
69     plt.grid(True)
70     plt.show()
71
72     print(f"Minimum found using n = {rate}: {minimizer}")
73

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75 # Graphing (non-convex case)
76 def graph_funcnt_non_convex(minimizer, trajectory, rate): 1 usage
77     x_vals = np.linspace(-3, stop: 3, num: 400)
78     y_vals = f_non_convex(x_vals)
79
80     plt.plot(*args: x_vals, y_vals, label='f(x) = x^4 - 4x^2 + 3')
81     traj_x = np.array(trajectory)
82     traj_y = f_non_convex(traj_x)
83
84     plt.scatter(traj_x, traj_y, color='red', marker='x',
85                label='Gradient Descent Path')
86
87     plt.xlabel('x')
88     plt.ylabel('f(x)')
89     plt.legend()
90     plt.title(f'Non-Convex Function - Gradient Descent (n = {rate})')
91     plt.grid(True)
92     plt.show()
93
94     print(f"Final point reached using n = {rate}: {minimizer}")
95

```

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97 # Main trials
98 if __name__ == "__main__":
99
100     print("\n(a) Small n → Convergence")
101     minimizer, trajectory = gradient_descent_convex(rate: 0.1, initial_x: 0.0, iters: 20)
102     graph_function_convex(minimizer, trajectory, rate: 0.1)
103
104     print("\n(b) Larger n → Faster convergence")
105     minimizer, trajectory = gradient_descent_convex(rate: 0.4, initial_x: 0.0, iters: 20)
106     graph_function_convex(minimizer, trajectory, rate: 0.4)
107
108     print("\n(c) Too large n → Divergence")
109     minimizer, trajectory = gradient_descent_convex(rate: 1.2, initial_x: 0.0, iters: 20)
110     graph_function_convex(minimizer, trajectory, rate: 1.2)
111
112     print("\n(d) Non-convex function → Local minima")
113     minimizer, trajectory = gradient_descent_non_convex(rate: 0.1, initial_x: 2.5, iters: 30)
114     graph_funct_non_convex(minimizer, trajectory, rate: 0.1)

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Run main x
C:\Users\Crina\PycharmProjects\analysis_hw5\.venv\Scripts\python.exe C:\Users\Crina\PycharmProjects\analysis_hw5\main.py

(a) Small n → Convergence
Minimum found using n = 0.1: 2.9654123548617948

(b) Larger n → Faster convergence
Minimum found using n = 0.4: 2.9999999999999685

(c) Too large n → Divergence
Minimum found using n = 1.2: -2507.0476627585413

(d) Non-convex function → Local minima
Final point reached using n = 0.1: -1.4142135729211092

Process finished with exit code 0

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Activate Windows
Go to Settings to activate Windows.



